CH 21 ELECTRIC FIELD AND ELECTRIC FORCES I

Intended Learning Outcomes – after this lecture you will learn:
1. Electric charge in real life
2. Electric force between point charges (Coulomb's law)
3. Electric field due to a point charge

Textbook Reference: 21.2 – 21.3

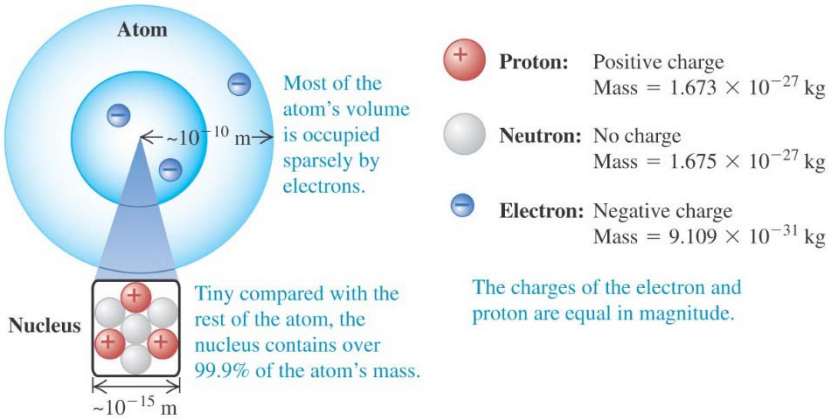
Why is E&M important?—most relevant to everyday life (from mechanical forces to biological processes) among the four fundamental interactions

The four fundamental forces:

	Relative strength	Range (meter)	Responsible for
Strong	10^{38}	10^{-15}	Binding quarks into hadrons, and neutrons and protons in nuclei
Electromagnetic	10^{36}	∞	
Weak	10^{25}	10^{-18}	Transforming neutron into proton in nuclear decay
Gravitation	1	∞	

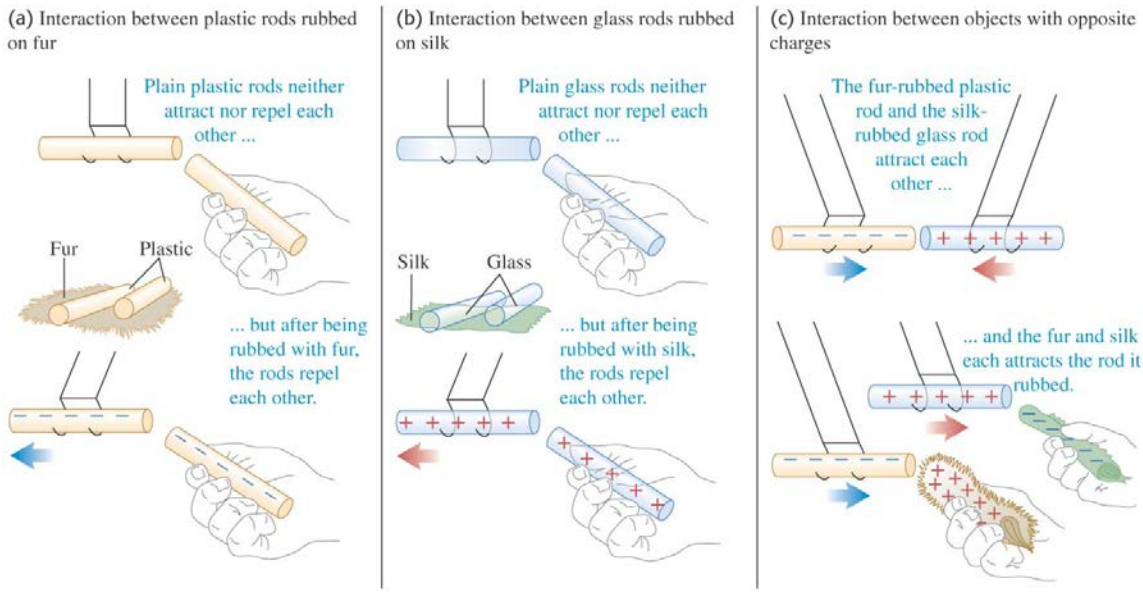
Some simple factors from high school physics:

Origin of charge from atoms

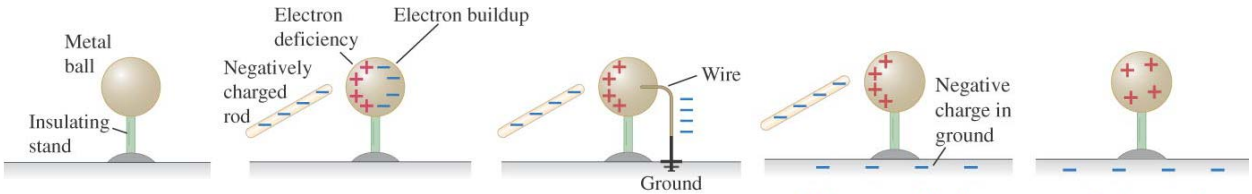


A conductor permits easy movement of charge through it, while an insulator does not

Charge by rubbing (insulators)



Charge by induction (conductors)



(a) Uncharged metal ball

(b) Negative charge on rod repels electrons, creating zones of negative and positive **induced charge**.

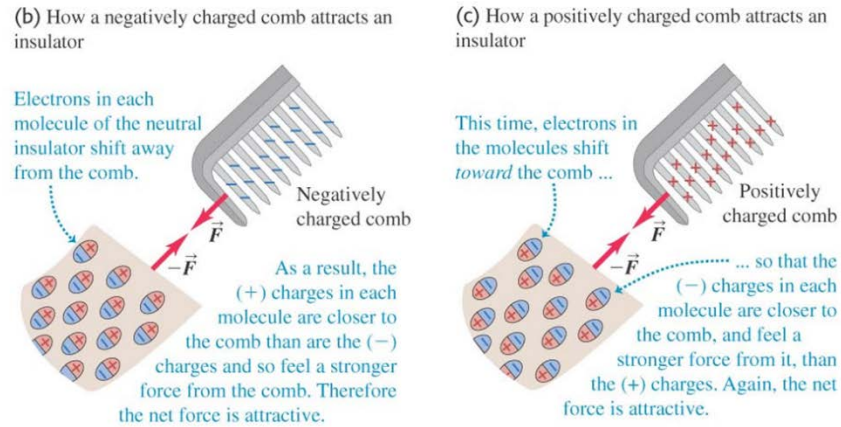
(c) Wire lets electron build-up (induced negative charge) flow into ground.

(d) Wire removed; ball now has only an electron-deficient region of positive charge.

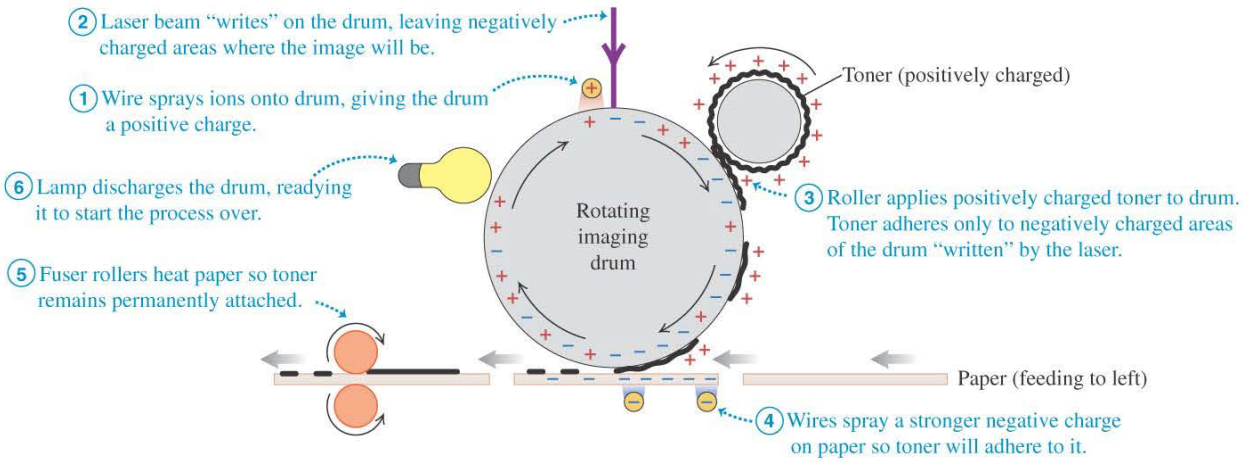
(e) Rod removed; electrons rearrange themselves, ball has overall electron deficiency (net positive charge).

Real Life Examples

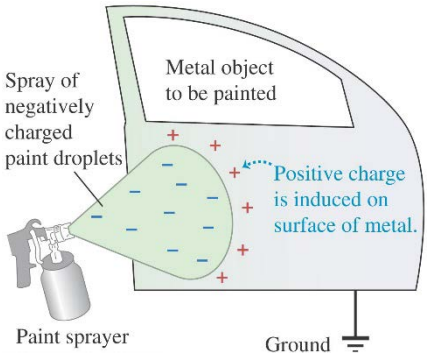
Combing hair in winter



Laser printing



Electrostatic painting



Minimizes overspray and gives a smooth finish

Question: Two metal balls, one charged and one neutral, hanging on insulating threads.
If they are close to each other but not touching, they will (attract / repel / ignore) each other.
Once they touch, they will (attract / repel / ignore) each other.
Answer: see inverted text on P. 712.

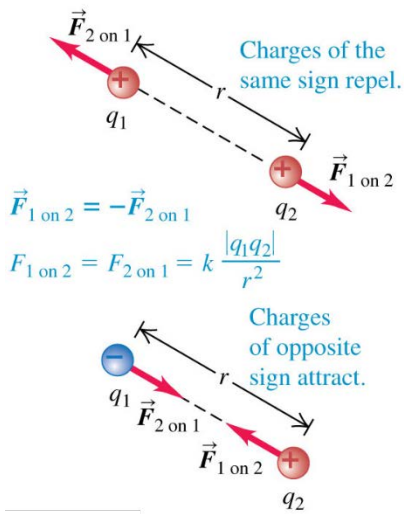
Two very important facts about charges:

1. **Principle of conservation of charge** – the algebraic sum of all the electric charges in any closed system (no charge can escape) is constant.
2. Electron charge is the fundamental (cannot be further divided) unit of charge
   $e = 1.602176565(35) \times 10^{-19}$ C

Demonstration: Van de Graaf generator

Raising hair Bouncing Balloon Lightning

Coulomb's Law



Magnitude of electric force between two point charges

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1q_2|}{r^2}$$

Vacuum permittivity

$$\epsilon_0 = 8.854 \times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2$$

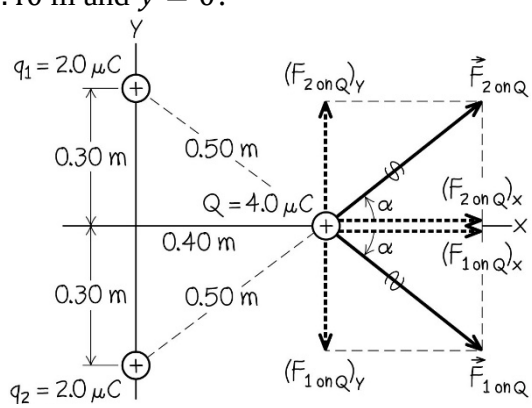
$$\frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2$$

⚠ Two 1 C charge at 1m apart exert a force of 9×10^9 N on each other! 1 C is a huge amount of electric charge!

Very much like gravitation, $F = Gm_1m_2/r^2$, but much stronger
E.g. two helium nuclei (He^{2+}, $m = 6.64 \times 10^{-27}$ kg and $q = 3.2 \times 10^{-19}$ C), ratio of electric to gravitational force is

$$\frac{F_e}{F_g} = \frac{\frac{1}{4\pi\epsilon_0}\frac{|q_1q_2|}{r^2}}{\frac{Gm_1m_2}{r^2}} = 3.1 \times 10^{35}$$

Example 21.4 P. 717 Vector Addition of Electric Forces in a Plane
Two equal positive charges $q_1 = q_2 = 2.0\ \mu$C are located at $x = 0$, $y = 0.30$ m and $x = 0$, $y = -0.30$ m respectively. Find the total electric force experienced by a charge $Q = 4.0\ \mu$C at $x = 0.40$ m and $y = 0$?



By symmetry, $F_y = 0$

$$F_x = 2\frac{1}{4\pi\epsilon_0}\frac{|Qq_1|}{r^2}\cos\alpha$$

$$= 2(9.0 \times 10^9\ \text{Nm}^2/\text{C}^2)$$

$$\times \frac{(4.0 \times 10^{-6}\ \text{C})(2.0 \times 10^{-6}\ \text{C})}{(0.50\ \text{m})^2}$$

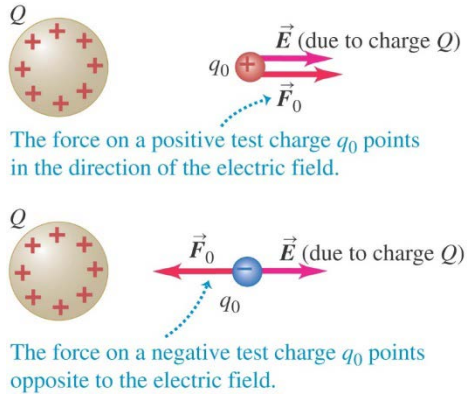
$$\times \frac{0.40\ \text{m}}{0.50\ \text{m}} = _____\text{N}$$

⚠ be careful about the precision!

Question: If $q_2 = -2.0\ \mu$C instead, the total force on Q will be (along x direction / along $-x$ direction / along y direction / along $-y$ direction / zero / none of these).
Answer: see inverted text on P. 718

⚠ Q feels the charge due to *other charges* (q_1 and q_2) only, never its own charge

⚠ If charge Q changed, just substitute a new value into above formula. Did q_1 and q_2 set up something independent of charge Q? Yes, they set up an **electric field**.



Electric field = electric force per unit charge

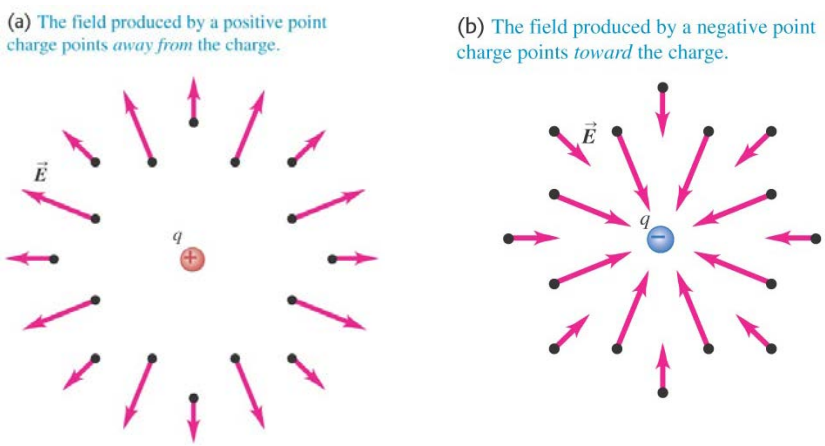
$$\vec{E} = \frac{\vec{F}_0}{q_0}$$

$\vec{F}_0$: Electric force on a test charge q_0 due to other charges

q_0: Value of test charge

Electric Field due to a Point Charge

Always radially outward (for +ve charge) or inward (for –ve charge)



Electric field due to a point charge

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r} \qquad (21.7)$$

q: Value of point charge

$\hat{r}$: Unit vector from point charge toward where field is measured

r: Distance from point charge to where field is measured

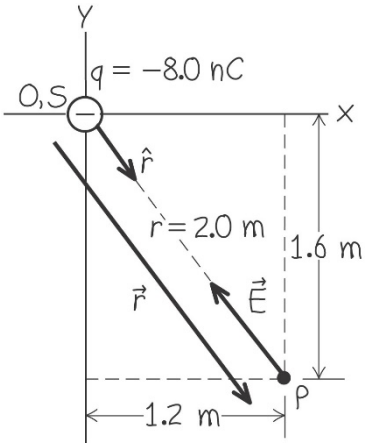
ϵ_0: Electric constant

⚠ $\hat{r}$ is <u>always</u> pointing radially outwards

⚠ q can be +ve/-ve.

Example 21.6 P. 721

A point charge $q = -0.8$ nC placed at the origin. The electric field at $x = 1.2$ m and $y = -1.6$ m is



EXECUTE: The distance from S to P is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.2\text{ m})^2 + (-1.6\text{ m})^2} = 2.0\text{ m}$$

The unit vector $\hat{\boldsymbol{r}}$ is then

$$\hat{\boldsymbol{r}} = \frac{\vec{\boldsymbol{r}}}{r} = \frac{x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}}{r}$$
$$= \frac{(1.2\text{ m})\hat{\boldsymbol{\imath}} + (-1.6\text{ m})\hat{\boldsymbol{\jmath}}}{2.0\text{ m}} = 0.60\hat{\boldsymbol{\imath}} - 0.80\hat{\boldsymbol{\jmath}}$$

Then, from Eq. (21.7),

$$\vec{\boldsymbol{E}} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$
$$= (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\frac{(-8.0 \times 10^{-9}\text{ C})}{(2.0\text{ m})^2}(0.60\hat{\boldsymbol{\imath}} - 0.80\hat{\boldsymbol{\jmath}})$$
$$= (-11\text{ N/C})\hat{\boldsymbol{\imath}} + (14\text{ N/C})\hat{\boldsymbol{\jmath}}$$

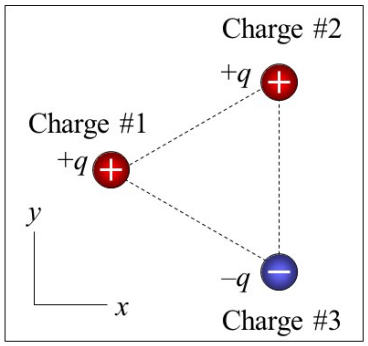
Question: A negative point charge moves along a circular orbit about a stationary positive point charge. The (magnitude / direction / magnitude and direction / neither magnitude nor direction) of the electric force on the negative charge will remain unchanged.

Answer: see inverted text on P. 722

Clicker Questions

Q21.3

Three point charges lie at the vertices of an equilateral triangle as shown. All three charges have the same magnitude, but charges #1 and #2 are positive ($+q$) and charge #3 is negative ($-q$). The net electric force that charges #2 and #3 exert on charge #1 is in



A. the $+x$-direction.

B. the $-x$-direction.

C. the $+y$-direction.

D. the $-y$-direction.

E. none of the above.

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Q21.5

A positive point charge $+Q$ is released from rest in an electric field. At any later time, the velocity of the point charge

A. is in the direction of the electric field at the position of the point charge.

B. is directly opposite the direction of the electric field at the position of the point charge.

C. is perpendicular to the direction of the electric field at the position of the point charge.

D. is zero.

E. Not enough information is given to decide.

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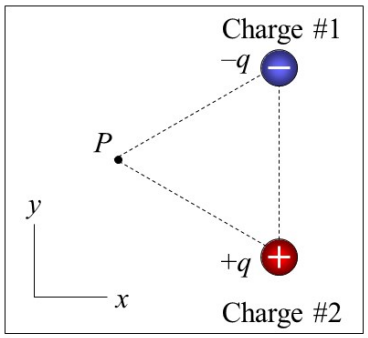
Q21.6

Two point charges and a point P lie at the vertices of an equilateral triangle as shown. Both point charges have the same magnitude q but opposite signs. There is nothing at point P. The net electric field that charges #1 and #2 produce at point P is in



A. the $+x$-direction.

B. the $-x$-direction.

C. the $+y$-direction.

D. the $-y$-direction.

E. none of the above.

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Ans: Q21.3) D, Q21.5) E, Q21.6) C